

Mehdi Yazdani-Jahromi

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Academic Services

Reviewer of Journal Briefings in Bioinformatics
 Reviewer of Computational and Structural Biotechnology Journal
 Reviewer of IEEE Transactions on Neural Networks and Learning Systems Journal
 Program Committee Member of AAAI Artificial Intelligence for Social Impact 2025 & 2026.
 Reviewer of Neurips 2025 (The Thirty-Ninth Annual Conference on Neural Information Processing Systems)

Education

PhD	University of Central Florida , Computer Science • Thesis: Advancing Drug Discovery with Structural and Representation Learning of Biological Systems Link • GPA: 3.91/4.0	Jan 2021 – Oct 2025
MS	University of Central Florida , Computer Science • GPA: 3.9/4.0	Jan 2021 – Aug 2023
MS	Sharif University of Technology , Industrial Engineering • GPA: 3.88/4.0	Sept 2017 – Dec 2019

Research Experience

DeepBio Scientific , Senior Research Scientist • Developed new benchmarking methods and protocol for agents in biology, which is the shortcoming of current protein generation/folding.	July 2026 – present
University of Central Florida , Graduate Research Assistant • Conducted advanced research in computational drug discovery, focusing on drug-target interaction and algorithmic fairness. • Developed and implemented machine learning models, including Transformers and Graph Neural Networks, for computer vision applications. • Collaborated on multiple interdisciplinary projects, contributing to the advancement of AI methodologies in drug discovery. • Published research findings in reputable journals and presented them at international conferences. • Utilized tools such as Pytorch, TensorFlow, and Scikit-learn to develop and evaluate innovative algorithms. • Engaged in data analysis and model optimization to enhance prediction accuracy and computational efficiency. • Assisted in mentoring undergraduate and graduate students and contributed to the academic community through collaborative efforts and knowledge-sharing sessions.	Orlando, FL Aug 2021 – Dec 2025

Microsoft, Research Intern, Bio-LLMs

- Built a novel agent orchestration system for planning, tool use, and judgment.
- Explored agentic AI on deidentified clinical datasets with end to end experiments and evaluation.
- Developed a RAG retrieval stack with hybrid dense and sparse search, reranking, and citation attribution.
- Contributed to the development of evaluation metrics and benchmarks for agentic AI systems.

Redmond, WA

June 2025 – Sept 2025

Johnson & Johnson (Janssen R&D), Data Science Intern, AI/ML for Drug Discovery

May 2024 – Nov 2024

- Developed and trained HELM (Hierarchical Encoding for mRNA Language Modeling), the first mRNA antibody language model, achieving up to 8% increase in prediction accuracy and enabling the generation of more diverse and biologically plausible sequences.
- Explored alternative attention architectures such as Mamba and Hyena, enhancing transformer model efficiency and effectiveness in processing mRNA sequences.
- Implemented large-scale distributed training on Kubernetes, resulting in significant reduction in training time for LLMs on extensive datasets, optimizing resource utilization and scalability.
- Collaborated with cross-functional teams to refine and deploy advanced LLM architectures, enhancing model accuracy and efficiency for large-scale data processing tasks.

Selected Publications

Equi-mRNA: Protein Translation Equivariant Encoding for mRNA Language Models

2025

Mehdi Yazdani-Jahromi, Ali Khodabandeh Yalabadi, Ozlem Ozmen Garibay

[10.48550/arXiv.2508.15103](https://arxiv.org/abs/2508.15103) (Neurips 2025)

BoKDiff: Best-of-K Diffusion Alignment for Target-Specific 3D Molecule Generation

2025

Ali Khodabandeh Yalabadi, **Mehdi Yazdani-Jahromi**, Ozlem Ozmen Garibay

[10.48550/arXiv.2501.15631](https://arxiv.org/abs/2501.15631) (Advances in Bioinformatics)

HELM: Hierarchical Encoding for mRNA Language Modeling

2024

Mehdi Yazdani-Jahromi, Mangal Prakash, Tommaso Mansi, Artem Moskalev, Rui Liao

[10.48550/arXiv.2410.12459](https://arxiv.org/abs/2410.12459) (ICLR 2025, Neurips 2024 Workshop on AI for New Drug Modalities)

Fair Bilevel Neural Network (FairBiNN): On Balancing fairness and accuracy via Stackelberg Equilibrium

Jan 2024

Mehdi Yazdani-Jahromi, Ali Khodabandeh Yalabadi, AmirArsalan Rajabi, Aida Tayebi, Ivan

Garibay, Ozlem Ozmen Garibay

[10.48550/arXiv.2410.16432](https://arxiv.org/abs/2410.16432) (Neurips 2024)

Learning Fair Representations: Mitigating Statistical Dependencies

Jan 2024

Aida Tayebi, **Mehdi Yazdani-Jahromi**, Ali Khodabandeh Yalabadi, Niloofar Yousefi, Ozlem Ozmen Garibay

[10.1007/978-3-031-60611-3_8](https://arxiv.org/abs/2303.03160) (HCII conference 2023 Oral Presentation)

FragXsiteDTI: an interpretable transformer-based model for drug-target interaction prediction

Jan 2024

Ali Khodabandeh Yalabadi, **Mehdi Yazdani-Jahromi**, Niloofar Yousefi, Aida Tayebi, Sina Abdidizaji, Ozlem Ozmen Garibay

[10.1007/978-1-0716-3989-4_5](https://arxiv.org/abs/2407.10716) (Recomb 2024 (Oral), Neurips 2023 Workshop on New Frontiers of AI for Drug Discovery and Development)

Controlling the misinformation diffusion in social media by the effect of different classes of agents Ali Khodabandeh Yalabadi, Mehdi Yazdani-Jahromi, Sina Abdidizaji, Ivan Garibay, Ozlem Ozmen Garibay 10.48550/arXiv.2401.11524 (The Computational Social Science Society of the Americas Annual Conference)	Jan 2023
Agent-Based Modeling of C. Difficile Spread in Hospitals: Assessing Contribution of High-Touch vs. Low-Touch Surfaces and Inoculations' Containment Impact Sina Abdidizaji, Ali Khodabandeh Yalabadi, Mehdi Yazdani-Jahromi, Ozlem Ozmen Garibay, Ivan Garibay 10.48550/arXiv.2401.11656 (The Computational Social Science Society of the Americas Annual Conference)	Jan 2023
Through a fair looking-glass: on mitigating bias in image datasets Amirarsalan Rajabi, Mehdi Yazdani-Jahromi , Ozlem Ozmen Garibay, Gita Sukthankar 10.1007/978-3-031-35891-3_27 (HCII conference 2023 (Oral), AAAI 2023 Workshop on Representation Learning for Responsible Human-Centric AI)	Jan 2023
BindingSiteAugmented DTA to enable A Next-Generation Pipeline for Interpretable Prediction Models in Drug-Repurposing Niloofar Yousefi, Mehdi Yazdani-Jahromi , Aida Tayebi, Elayaraja Kolanthai, Craig J Neal, Tanumoy Banerjee, Agnivo Gosai, Ganesh Balasubramanian, Sudipta Seal, Ozlem Ozmen Garibay 10.1093/bib/bbad136 (Briefings in Bioinformatics)	Jan 2023
AttentionSiteDTI: Attention Based Model for Predicting Drug-Target Interaction Using 3D Structure of Protein Binding Sites Mehdi Yazdani-Jahromi , Niloofar Yousefi, Aida Tayebi, Elayaraja Kolanthai, Craig J Neal, Sudipta Seal, Ozlem Ozmen Garibay 10.1093/bib/bbac272 (Briefings in Bioinformatics)	Jan 2022
UnbiasedDTI: Mitigating Real-World Bias of Drug-Target Interaction Prediction Aida Tayebi, Niloofar Yousefi, Mehdi Yazdani-Jahromi , Elayaraja Kolanthai, Craig J Neal, Sudipta Seal, Ozlem Ozmen Garibay 10.3390/molecules27092980 (MDPI Molecules)	Jan 2022

Skills

Programming Languages: Python, Dart, JavaScript, C++, C#, SQL, Solidity, MATLAB

Python Packages: Pytorch, Numpy, Pandas, Keras, TensorFlow, Scikit-learn, Scipy, Networkx, iGraph, dgl, matplotlib, seaborn

JavaScript Frameworks: Vue.js, Electron, NativeScript

Tools: Kubernetes, Docker, Git, HTML, CSS, LATEX, AWS, Azure, Slurm

Quantitative Research: Machine Learning Methods, Artificial Intelligence, Large Language Models, Generative AI, Genomics Sequence Modeling, Drug Discovery, Graph Neural Networks, Transformers, Recurrent Neural Networks, LSTMs, Mathematical Optimization, Mathematical Modeling

Soft Skills: Research team coordination, Student mentoring, Technical writing, Conference presentations, Resource allocation, Strategic planning, Analytical thinking, Complex problem resolution, International team collaboration, Multi-cultural communication, Self-directed learning, Professional networking

Summary

AI research scientist and PhD in Computer Science specializing in representation learning, with primary applications in computational biology and drug discovery. Industry experience at Johnson and Johnson and Microsoft, where I developed language models for mRNA and machine learning systems for large scale biomedical data. Strong programming background in Python, and JavaScript, with deep experience in PyTorch, Transformers and graph neural networks. Published in NeurIPS, ICLR and Briefings in Bioinformatics, with work spanning mRNA modeling, drug target interaction prediction and algorithmic fairness.